ME-295, Lab Report 1

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Topic of Interest

All space structures that are designed and envisaged for orbit or space exploration must overcome one significant hurdle: launch from Earth. The tight volume constraints of launch vehicles mean that any structure larger than a few meters wide must be either built or deployed in space [7]. Even though in-space manufacturing and assembly are receiving growing attention [8], deployable structures remain the standard technique to unfold large solar arrays, antennas, and science instruments [7]. Of particular interest are high-strain composite materials [1] that can undergo significant deformation, with radii of curvature on the order of tens of millimeters, for use as hinges or collapsible masts [2]. These collapsible masts, also known as booms or longerons, are relatively thin and long structures with a uniform cross-section throughout their length. The key feature of this cross-section is that it can be flattened by pressing against a flat surface, which allows the boom to then be coiled about a cylindrical hub so that multi-meter-long booms can be stowed in a cylindrical volume of roughly 100 mm diameter by 100 mm height [2], [9].

After almost a decade of rapid advancements in new designs for these collapsible booms, many different cross-sections have been considered [3], each with advantages and disadvantages depending on mission requirements. Some studies have optimized the geometry of these booms using machine-learning techniques. These works trained neural networks on FEA results of a single boom geometry (e.g., the omega boom), varying curvature radii and angles to determine which parameters offer the highest buckling strength, stiffness, or resistance to failure during coiling [4], [5]. These methods can speed up the optimization of a few geometrical parameters, but they do not attempt to also optimize for the material properties of the fiber-reinforced composite that makes up the thin shells of these structures, nor do they attempt to compare completely different cross-sections automatically.

This gap in the literature is likely due to the large effort required to write code that automatically generates different boom cross-sections and adapts the mesh density for each geometry. In addition, varying the material properties can often cause convergence issues [6] arising from changes in how buckles develop in the deployable masts. Yet, being able to quickly identify the best cross-section and material layup for a deployable boom for a given mission requirement is highly desirable to speed up the design process of future space missions.

Project Statement

The objective of the current project is to develop an AI agent [11] that can use the OpenSeesPy open-source finite element library for Python [10] to autonomously create the geometry and mesh of a deployable mast. It should also calculate the orthotropic stiffness matrix for the material and assign it to the model, based on the known properties of individual plies for different carbon-fiber/matrix combinations. The agent should then run a series of analyses, ranging from bending and twisting to determine the stiffnesses of the boom to coiling to determine the internal stresses and the likelihood of failure of the composite [12], in order to obtain metrics for comparison against other booms. Ultimately, the goal is that, after sufficient training, the agent will be able to receive mission parameters, such as packaged volume and maximum loads, and determine the lightest or most failure-resistant deployable boom design.

Current Progress

At this time, our GitHub repository has been created, and both team members can make changes to files in the repository directly from the AWS sandbox. Claude Code has been installed on the machines, and we have made first attempts at having the agent create and modify files. Python is not yet installed in the workspace, so the code written by the agent has not yet been tested. However, the next steps in the project involve prompting the agent and manually checking its ability to:

- create simple finite element models of beams
- provide different material properties to these beams based on varying composite layups
- apply varying loads and boundary conditions and check how the agent gathers and analyzes the results
- test the limits of OpenSeesPy with the current capabilities of the AWS sandbox, to determine how complex an FEA model can be while still running in a reasonable amount of time
- test, through the agent, the different types of simulations we are interested in, such as coiling or buckling analyses
- move on to more complex geometries and adapt the same approach to see if the agent can run these simulations and address problems on its own

AI disclosure

Claude code was used to search for additional literature and embed it in the text of the report. In addition, it read the text and fixed any grammar/improved the sentence srtucture.

References

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AI suggested References

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